

Average Access Is Not Inclusion: Sharp Transport Bounds for Multidimensional Digital Capability Gaps

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Abstract

Separate rates of phone ownership, internet use and account ownership can leave joint inclusion uncertain. We use finite transport and classical moment bounds to link completion technologies to their data requirements. With a fixed fee per recipient, publishing the three-way inclusion share identifies baseline completion cost. If the release menu permits one exact pairwise overlap at equal cost, a pair excluding a dimension with the highest marginal minimizes the worst remaining interval. In the 2024 wave of Global Findex 2025, marginal rates leave a median inclusion interval of 29.7 percentage points across 93 low- and middle-income economies. The marginal-based rule reduces this to 11.9 points and selects the most informative observed pair in every economy. All three pairwise overlaps reduce the median to 2.1 points. These releases also resolve budget and group comparisons under normalized costs. A stricter device requirement and digital-payment cross-tabulations clarify the bundle's empirical scope. The results condition on survey estimates; population inference and monetary budgets require survey-design and programme evidence.

Keywords: digital inclusion; partial identification; optimal transport; multidimensional deprivation; policy costs; dependence uncertainty.

JEL classification: C61, D63, I32, O15, O33.

1 Introduction

How many adults possess a phone, use the internet and have a financial account? What would it cost to extend all three conditions to everyone? Coverage statistics typically report each separately. Joint inclusion depends on whether the same adults meet all three conditions. Their overlap also determines how many people need assistance and, under many delivery technologies, the cost of closing the remaining gaps.

Suppose the three coverage rates are 0.85, 0.70 and 0.65. The share possessing all three can range from 0.20 to 0.65. If each missing capability costs one unit to provide and each recipient costs one additional unit to enrol, universal completion costs between 1.15 and 1.60 units per adult. Both endpoints are attainable. Concentrated deficits require fewer enrolments, so the same coverage rates can imply different budgets.

This paper measures the information lost when joint survey records are summarized by marginal coverage rates. We represent an intervention as a sequence of capability upgrades and calculate the cheapest route from each initial profile to a specified standard. Without

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shared capacity constraints, total cost is the mean of these individual completion costs. Classical moment bounds then give its smallest and largest values across compatible joint distributions. This connects a policy’s data requirements to its delivery technology.

We apply results on moment identification and uniform approximation to determine when marginals suffice and how much uncertainty can remain. Under a fixed fee per recipient, cost uncertainty is proportional to uncertainty about the complete-bundle share. Publishing that share directly identifies both quantities. We also solve a narrower design problem: if one exact pairwise overlap may be released at equal cost, choosing the pair that excludes a dimension with the highest marginal minimizes the worst remaining interval. The rule uses only marginal rates.

The application uses the 2024 wave of Global Findex 2025 (Klapper et al., 2025). Among 93 low- and middle-income economies, marginal rates leave a median interval of 29.7 percentage points for the share with all three capabilities. The marginal-based rule reduces this to 11.9 points and selects the most informative observed pair in every economy. It chooses internet–account in the 85 economies where phone ownership is the highest marginal. Releasing all three pairs reduces the median width to 2.1 points; the direct three-way share removes the remaining identification uncertainty. We also measure how these releases resolve budget and group comparisons. The exercise holds records and weights fixed, with costs in normalized units. Population inference and monetary budgets require survey-design and programme evidence, respectively.

1.1 Related literature

The paper draws on multidimensional poverty measurement. Bourguignon and Chakravarty (2003) and Alkire and Foster (2011) develop methods for identifying multidimensional poverty and aggregating individual deprivations, and Ferreira and Lugo (2013) explain why marginal dashboards miss their overlap. Policy research also incorporates this dependence: Duclos et al. (2018) study multidimensional poverty targeting, while Santos et al. (2026) assess feasible poverty reduction and minimum spending using deprivation costs. We ask which features of the joint distribution are required to evaluate a specified completion technology. Welfare comparisons additionally require a valuation of the capability standard and its benefits.

The probability bounds and duality have established antecedents. Hailperin (1965) studies best possible probabilities of logical combinations of events, and Fréchet–Hoeffding bounds give our baseline intersection interval (Nelsen, 2006). Kellerer (1984), Bertsimas and Popescu (2005) and Bartl et al. (2022) develop marginal duality, optimization under moment restrictions and bounds with additional dependence information. The approximation result below applies the moment-matching principle whose polynomial form appears in Han et al. (2018, Lemma 25). Our finite-state arguments specialize these principles to capability completion. Additional valid moments refine the identified set as in the partial-identification literature (Manski, 2003).

Optimal transport describes how a technology moves a population across capability states (Galichon, 2016). Because we leave the destination distribution free within the target set, the baseline problem separates across recipients; capacity constraints remove this separability. Our policy analysis also relates to moment-based distributionally robust optimization (Delage and Ye, 2010). We distinguish a common policy chosen before the population’s composition is known from allocations that adapt afterward.

Joint digital-access analysis also has a precedent in the Global Findex 2017 report, which combines phone ownership and internet access among adults without an account (Demirgüç-Kunt et al., 2018, Figure 6.3). A recent preprint studies digital capabilities, device security and financial participation using Findex data (Echeverría Mayorga et al., 2026). Our application quantifies the loss from reporting only marginals and compares explicit release menus. It links

the choice of statistics to completion costs, evaluates a prospective pair-selection rule, and measures the budget and group comparisons that additional information resolves.

2 Capability states and directed policy transport

Let $\mathcal{S} = \{0, 1\}^d$, where $x_j = 1$ denotes possession of capability j . Write $x \preceq y$ if $x_j \leq y_j$ for every j . The policy target is a nonempty, upward-closed set $A \subseteq \mathcal{S}$: $x \in A$ and $x \preceq y$ imply $y \in A$. Examples include the complete bundle $A = \{\mathbf{1}\}$, a threshold $A = \{x : \sum_j x_j \geq k\}$, or a set requiring only certain essential capabilities. The policymaker specifies this standard.

An intervention graph G has directed edges (x, y) only when $x \preceq y$. An edge adds one or more capabilities at a nonnegative cost. Let $d_G(x, y)$ be the minimum path cost, with $d_G(x, x) = 0$ and infinity for unreachable destinations. Assume every admissible initial state can reach A at finite cost. Its minimum completion cost is

$$h_A(x) = \min_{y \in A} d_G(x, y). \quad (1)$$

Thus $h_A(x) = 0$ for people already in A . Costs may vary across initial profiles and are the same for people sharing a profile. The baseline has no shared capacity constraints.

For an initial population distribution μ , define the universal-completion cost

$$T_A(\mu) = \min_{\pi \geq 0} \left\{ \sum_{x, y} d_G(x, y) \pi_{xy} : \sum_y \pi_{xy} = \mu_x, \quad \pi_{xy} = 0 \text{ if } y \notin A \right\}. \quad (2)$$

Only finite-cost arcs may carry mass. The destination distribution is unrestricted within A .

Proposition 1 (Reduction to statewise completion costs). *Under the finite-reachability assumption,*

$$T_A(\mu) = \sum_x h_A(x) \mu_x = \mathbb{E}_\mu h_A(X). \quad (3)$$

An optimal transport sends all mass at each x to a least-cost reachable state in A .

The problem separates across people. The intervention graph determines h_A ; optimizing its expectation over distributions with fixed marginals gives a finite multi-marginal transport problem. Uncertainty about joint capabilities affects total cost only through nonadditivity in h_A .

2.1 Upgrade costs and a fixed delivery fee

For universal completion into the full-bundle target $A = \{\mathbf{1}\}$, let $k_j \geq 0$ be the cost of adding capability j , and let $f \geq 0$ be a fixed delivery cost for each initially incomplete person. A direct upgrade from x to a strictly higher y costs $f + \sum_j k_j(y_j - x_j)$. Including these bundle edges gives

$$h_{f, k}(x) = f \mathbf{1}\{x \neq \mathbf{1}\} + \sum_j k_j(1 - x_j). \quad (4)$$

A direct upgrade is cheapest because separate visits repeat the fixed fee. For $f = 0$, expected cost is $\sum_j k_j(1 - p_j)$, regardless of dependence. For $f > 0$, it also depends on how many distinct people need assistance.

The fee f represents enrolment or delivery, and k_j represents an upgrade cost. Equation (4) specifies a policy scenario in which these expenditures deliver the required capabilities. Empirical calibration requires programme cost and effectiveness data.

3 Sharp identification from marginal information

Suppose published data identify $p_j = \mathbb{E}X_j$. The unrestricted Fréchet class is

$$\mathcal{F}(p) = \left\{ \mu \geq 0 : \sum_x \mu_x = 1, \quad \sum_x x_j \mu_x = p_j \quad (j = 1, \dots, d) \right\}. \quad (5)$$

Every $p \in [0, 1]^d$ is feasible. The sharp identification endpoints are

$$\underline{T}(p) = \min_{\mu \in \mathcal{F}(p)} h^\top \mu, \quad \overline{T}(p) = \max_{\mu \in \mathcal{F}(p)} h^\top \mu, \quad W_h(p) = \overline{T}(p) - \underline{T}(p), \quad (6)$$

where $h = h_A$. For the specified target and technology, the lower endpoint is the smallest budget consistent with the data. The upper endpoint covers every compatible distribution, provided the programme can deliver each person's least-cost upgrade.

Theorem 2 (Sharpness and affine dual certificates). *For finite h , both endpoints are attained and the identified set is the entire interval $[\underline{T}(p), \overline{T}(p)]$. Moreover,*

$$\underline{T}(p) = \max_{\alpha, \beta} \left\{ \alpha + \beta^\top p : \alpha + \beta^\top x \leq h(x) \quad \forall x \right\}, \quad (7)$$

$$\overline{T}(p) = \min_{\alpha, \beta} \left\{ \alpha + \beta^\top p : \alpha + \beta^\top x \geq h(x) \quad \forall x \right\}. \quad (8)$$

There are extremizing distributions supported on at most $d + 1$ states. The lower endpoint is convex in p , and the upper endpoint is concave in p .

The dual bounds h by affine functions whose expectations are known from the marginals. At an optimum, mass lies only on states where the relevant affine bound equals h . The dual coefficients, which may be nonunique, describe the sensitivity of the bound to the reported moments under a fixed technology. Causal effects of interventions require separate identification.

The standard row-space criterion yields the following characterization.

Theorem 3 (Exactly when the marginals suffice). *Fix $p \in (0, 1)^d$. The interval in (6) is a singleton if and only if there exist $a \in \mathbb{R}$ and $b \in \mathbb{R}^d$ such that*

$$h(x) = a + b^\top x \quad \text{for every } x \in \mathcal{S}. \quad (9)$$

Equivalently, point identification holds for every feasible marginal vector if and only if h is affine on the cube. If some p_j equal zero or one, the fixed- p statement applies to the surviving face only.

The relevant object is the minimum completion cost h . Complicated delivery routes can produce an affine h ; additive edge costs can produce a nonaffine h when the target permits a choice among capabilities.

The same argument applies to restricted support and additional moments. Let $b(x)$ collect a constant and the observed functions, let $m = \mathbb{E}b(X)$, and define the active support $S^*(m) = \{x : \mu_x > 0 \text{ for some feasible } \mu\}$. Then $\mathbb{E}h$ is point identified if and only if h lies in the span of the observed functions on $S^*(m)$. Only states that can carry positive mass matter.

3.1 A quantitative measure of nonadditivity

Define the best uniform affine approximation error

$$\delta(h) = \inf_{a, b} \max_{x \in \mathcal{S}} |h(x) - a - b^\top x|. \quad (10)$$

This measures the departure from a cost function identified by the marginals. The following result is a finite-state form of moment-matching and uniform-approximation duality; Han et al. (2018, Lemma 25) give its polynomial-moment form on an interval.

Theorem 4 (Maximal ambiguity and affine approximation). *For finite h on the cube,*

$$W_h(p) \leq 2\delta(h) \quad \text{for every } p, \quad \max_{p \in [0,1]^d} W_h(p) = 2\delta(h). \quad (11)$$

The same equality holds with the span of any collection of observed functions that includes a constant, maximizing over its feasible moment vectors.

An approximately affine cost function therefore leaves little ambiguity at any marginal vector. The bound is attained at some vector, though the interval may be narrower at a particular economy's marginals.

4 Monotonicity, stability and the value of information

4.1 When better capabilities reduce completion costs

Assumption 1 (Monotone completion technology). The statewise minimum cost is decreasing in the coordinate order: if $x \preceq x'$, then $h_A(x') \leq h_A(x)$.

Nonnegative edge costs are not enough. On the two-dimensional cube with target 11, suppose $00 \rightarrow 01$ and $01 \rightarrow 11$ each cost one, while $00 \rightarrow 10$ costs one and $10 \rightarrow 11$ costs ten. Then $h(00) = 2 < h(10) = 10$: acquiring the first capability removes access to the cheap route.

A sufficient condition is that any programme available from x can also be used from $x' \succeq x$ at weakly lower cost, with an endpoint at least as high. On a full single-coordinate graph, this holds if adding a capability costs no more when other capabilities are present; already-owned coordinates can be skipped. Specification (4) satisfies Assumption 1.

Proposition 5 (Dominance and marginal monotonicity). *Under Assumption 1, if $\mu'(U) \geq \mu(U)$ for every upward-closed set U , then $T_A(\mu') \leq T_A(\mu)$. On the unrestricted cube, if $p' \geq p$ coordinatewise, then*

$$\underline{T}(p') \leq \underline{T}(p), \quad \overline{T}(p') \leq \overline{T}(p). \quad (12)$$

The first statement compares joint distributions; the second compares bounds over moment classes. Ordered marginal means alone do not order the costs of two given joint distributions.

4.2 Stability with respect to published marginals

Let $L_j = \max_{x: x_j=0} |h(x + e_j) - h(x)|$. These constants are finite on the cube.

Proposition 6 (Global marginal stability). *For $p, q \in [0, 1]^d$,*

$$|\underline{T}(p) - \underline{T}(q)| \leq \sum_j L_j |p_j - q_j|, \quad |\overline{T}(p) - \overline{T}(q)| \leq \sum_j L_j |p_j - q_j|. \quad (13)$$

These inequalities bound the effect of rounding or measurement error in the marginals. They require unrestricted support: under structural restrictions, a perturbed marginal vector may be infeasible.

4.3 Cross-tabulations and interaction order

Let $\mathcal{I}$ be a collection of subsets of coordinates and suppose $m_J = \mathbb{E} \prod_{j \in J} X_j$ is reported for $J \in \mathcal{I}$, in addition to the marginals. Adding these restrictions shrinks the feasible distribution

set. Therefore the lower endpoint weakly increases and the upper endpoint weakly decreases. If the observed functions span h on the active support, the interval collapses.

Every function on a binary cube has a unique multilinear expansion

$$h(x) = \sum_{J \subseteq \{1, \dots, d\}} \theta_J \prod_{j \in J} x_j. \quad (14)$$

If the information consists of all joint moments through order r and a fully supported distribution is compatible with these moments, then $\mathbb{E}h$ is point identified if and only if $\theta_J = 0$ for every $|J| > r$. In particular, all pairwise tables need not identify a three-way completion event. Pairwise intersections equal to $1/4$ and marginal probabilities equal to $1/2$ are shared by the uniform distributions on $\{000, 011, 101, 110\}$ and on $\{001, 010, 100, 111\}$. Their full-bundle probabilities are zero and $1/4$, respectively.

For a cross-tabulation with observed value q , the gain in identification is $W(m) - W(m, q)$. The empirical release comparisons in Section 10 use this realized gain. Ranking releases before q is known requires an expected, worst-case or decision-based criterion.

5 Closed-form completion budgets

For the full-bundle target under (4), write $D_j = 1 - p_j$, $K = \sum_j k_j D_j$, and $s = \Pr(X = \mathbf{1})$. Then

$$T_{f,k}(\mu) = K + f(1 - s), \quad \max\{0, \sum_j p_j - (d - 1)\} \leq s \leq \min_j p_j. \quad (15)$$

Corollary 7 (Fixed delivery costs). *Under unrestricted dependence,*

$$\underline{T}_{f,k}(p) = K + f \max_j D_j, \quad (16)$$

$$\bar{T}_{f,k}(p) = K + f \min\{1, \sum_j D_j\}, \quad (17)$$

$$W_{f,k}(p) = f \left[\min\{1, \sum_j D_j\} - \max_j D_j \right]. \quad (18)$$

All bounds are attained.

The additive component K affects the level of costs, while f scales the interval's width: $W_{f,k} = fW_s$, where W_s is the inclusion-interval width. Thus the baseline cost bounds are an affine transformation of classical intersection bounds. For $f > 0$, releases have the same ranking for both quantities and, when the initial interval has positive width, the same proportional reduction. The cost interpretation translates overlap uncertainty into resource requirements under the specified technology.

Nested deprivation events attain the lower endpoint: everyone with a less common deficit also has the most common one. Their union has probability $\max_j D_j$. Spreading deficits across as many people as possible attains the upper endpoint, with union probability $\min\{1, \sum_j D_j\}$. Dispersion raises delivery costs because it increases the number of recipients.

Example 1 (Same marginals, different costs). Take $(p_1, p_2, p_3) = (0.85, 0.70, 0.65)$ and $f = k_1 = k_2 = k_3 = 1$. Distribution μ^L places mass $(0.15, 0.15, 0.05, 0.65)$ on $(000, 100, 110, 111)$. Distribution μ^U places mass $(0.15, 0.30, 0.35, 0.20)$ on $(011, 101, 110, 111)$. Both have the specified marginals. Their completion costs are 1.15 and 1.60, respectively. The additive cost is 0.80 in both cases, while enrolment reaches 35% and 80% of the population. Imposing independence gives $s = 0.38675$ and cost 1.41325, an interior point selected by this additional dependence assumption.

The formulas also give the dual affine bounds. The lower bound uses a coordinate attaining $\max_j D_j$; the upper bound depends on whether $\sum_j D_j$ is below or above one. Kinks and ties can make the associated slopes nonunique.

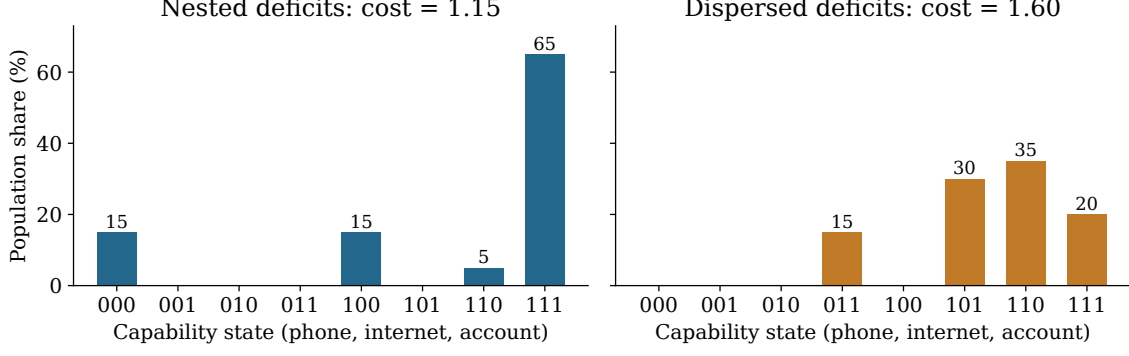


Figure 1: Identical marginal access, different completion costs. Bars show the constructed state probabilities in Example 1, with coordinates ordered as phone, internet and account. The two distributions attain the sharp cost endpoints.

6 Budgeted coverage, adaptation and group comparisons

6.1 Covering a specified share

Let q_x be the mass selected from state x , including people already in the target at zero cost. Allowing selection of part of a state group, the least cost of including a share $\rho \in [0, 1]$ is

$$C_\rho(\mu) = \min_{0 \leq q \leq \mu} \left\{ \sum_x h(x) q_x : \sum_x q_x \geq \rho \right\}. \quad (19)$$

Fractional selection can be implemented by randomization in a large population. Small programmes with indivisible resources may require an integer formulation.

Proposition 8 (Partial-coverage dual and sharp ambiguity bounds). *Let $H = \{0\} \cup \{h(x) : x \in \mathcal{S}\}$. For every μ ,*

$$C_\rho(\mu) = \max_{t \in H} \{t\rho - \mathbb{E}_\mu(t - h(X))_+\}. \quad (20)$$

Its sharp upper bound over $\mathcal{F}(p)$ is

$$\overline{C}_\rho(p) = \max_{t \in H} \left\{ t\rho - \min_{\mu \in \mathcal{F}(p)} \mathbb{E}_\mu(t - h(X))_+ \right\}. \quad (21)$$

The sharp lower bound is the joint linear programme in (μ, q) with $\mu \in \mathcal{F}(p)$ and the constraints in (19). Both bounds are attained.

The upper bound has order $\max_\mu \min_q$: the programme chooses the cheapest allocation for each possible population. Joint maximization over (μ, q) would allow deliberately costly targeting. Equation (21) evaluates the adaptive bound through finitely many moment programmes.

For budget $B \geq 0$, define $R(B; \mu) = \max\{\rho : C_\rho(\mu) \leq B\}$. The largest adaptively guaranteed coverage share is

$$R_{\text{ad}}(B; p) = \sup\{\rho : \overline{C}_\rho(p) \leq B\} = \inf_{\mu \in \mathcal{F}(p)} R(B; \mu). \quad (22)$$

The guarantee assumes that the programme observes individual needs and the eligible pool's composition before allocating treatment. Screening is free or funded separately; f covers delivery to selected recipients. A rule that fixes state-specific treatment fractions before learning the composition can require a larger budget (Appendix B).

6.2 Group-specific information and disparities

Let g index a mutually exclusive and exhaustive partition with known shares w_g . Suppose marginal vectors p_g are available and no other constraints link group distributions. Then

$$\min T = \sum_g w_g \underline{T}(p_g), \quad \max T = \sum_g w_g \bar{T}(p_g). \quad (23)$$

Choosing each group's extremizer attains these endpoints. Group information weakly narrows the interval relative to pooled marginals $p = \sum_g w_g p_g$.

For two groups with no cross-group restrictions, the sharp per-person cost-gap interval is

$$T_g - T_{g'} \in [\underline{T}(p_g) - \bar{T}(p_{g'}), \bar{T}(p_g) - \underline{T}(p_{g'})]. \quad (24)$$

A positive lower endpoint means that group g costs more to serve under every compatible dependence pattern. An interval containing zero leaves this strict ordering unresolved.

Gender, residence and income tables describe separate partitions. Combining them requires joint group data to account for overlapping membership and identify groups such as poor rural women.

6.3 Capacity and equity restrictions

When delivery routes compete for shared resources, the statewise reduction no longer suffices. Let a index complete admissible routes from x into A , with cost c_{gxa} and resource requirements $r_{gxa\ell}$. A population-mass allocation z_{gxa} satisfies

$$\sum_a z_{gxa} \leq w_g \mu_g(x), \quad \sum_{g,x,a} r_{gxa\ell} z_{gxa} \leq K_\ell, \quad \sum_{x,a} z_{gxa} \geq w_g \rho_g. \quad (25)$$

Minimizing $\sum c_{gxa} z_{gxa}$ gives the cost of meeting group coverage floors ρ_g subject to capacity limits. Already acceptable states have a zero-cost route. Insufficient capacity can make the problem infeasible. When the constrained and relaxed problems are both feasible, the cost increase is nonnegative. Under standard local sensitivity conditions, dual multipliers measure the cost of tightening a coverage floor or the saving from relaxing a resource constraint.

Estimating (25) requires route costs, capacities and coverage requirements beyond the Findex data.

7 Choosing an additional statistic

Consider an agency that holds joint survey records and an analyst who observes only the released summaries. The baseline target has a simple sufficient release. Given the marginals, publishing $s = \Pr(X = 111)$ identifies both joint inclusion and $T_{f,k} = K + f(1 - s)$. An exact release of this one statistic therefore gives zero remaining width. More generally, publishing the mean of a specified cost function identifies that cost directly. Such targeted releases are useful benchmarks; pairwise tables support a wider class of uses, including all quadratic cost functions on the binary cube. The available release menu and intended uses therefore define the design problem.

Consider a restricted menu containing the three pairwise overlaps. An agency selects one fixed pair using the marginals, before observing the overlap to be released. Each candidate is reported exactly and has the same release cost. The objective is to minimize the largest remaining inclusion interval over compatible populations. This is a specified design scenario: an application would need to establish its reporting constraints and release costs. Sampling error, privacy noise and other policy uses require additional criteria.

Proposition 9 (Single-pair bounds and a minimax release). *Let $\{i, j, k\} = \{1, 2, 3\}$, $p \in [0, 1]^3$, and $q_{ij} = \Pr(X_i = X_j = 1)$. Define $\ell_{ij} = \max\{0, p_i + p_j - 1\}$ and $u_{ij} = \min\{p_i, p_j\}$. For every $q_{ij} \in [\ell_{ij}, u_{ij}]$, the sharp inclusion interval is*

$$\begin{aligned} \max\{0, q_{ij} + p_k - 1\} &\leq s \leq \min\{q_{ij}, p_k\}, \\ w_{ij}(p, q_{ij}) &= \min\{q_{ij}, 1 - q_{ij}, p_k, 1 - p_k\}. \end{aligned} \quad (26)$$

Write $p_{\min} = \min_j p_j$ and $p_{\max} = \max_j p_j$. The largest remaining width for this pair is

$$R_{ij}(p) = \max_{q \in [\ell_{ij}, u_{ij}]} w_{ij}(p, q) = \min\{p_{\min}, 1 - p_k, 2 - p_i - p_j\}, \quad (27)$$

$$\min_{\{i, j\}} R_{ij}(p) = \min\{p_{\min}, 1 - p_{\max}\}. \quad (28)$$

A pair omitting any coordinate with the largest marginal attains the minimum in (28). Other pairs may also attain it.

The proof in Appendix C applies the classical intersection bounds and verifies sharpness while retaining all three marginals.

Under the fixed-fee technology, each width in the proposition is multiplied by f to obtain the corresponding cost width. At $f = 0$, the marginals already identify cost. The prospective rule uses only the marginal rates. After overlaps become available, their realized values determine the narrowest interval through (26). The two choices can differ, as the appendix illustrates. In particular, an internet-account release leaves width at most $1 - p_{\text{phone}}$. We compare these implications with the observed release rankings.

8 Data and empirical implementation

8.1 Data and capability measures

We use the published, survey-weighted estimates from the 2024 wave of Global Findex 2025 (Klapper et al., 2025). The surveys cover the civilian, noninstitutionalized population aged 15 and above and report marginal rates by economy and demographic group. Most interviews took place in 2024; fieldwork in Eswatini continued into January 2025.

Our three dimensions are personal mobile-phone ownership, internet use in the preceding three months, and financial-account ownership (source columns `con1`, `internet`, and `account_t_d`). Phone ownership includes basic phones and smartphones used for personal calls. Internet use combines affirmative responses to the seven-day question and its three-month follow-up. Account ownership includes individual or joint accounts at banks or similar institutions, and personal use of mobile-money services during the preceding year. All three rates use the same denominator: all adults in the economy or subgroup.

These indicators define a specific capability bundle. Literacy, service quality, independent account control, regular online transactions, and safety require additional measures. The bundle includes accounts without digital access and basic phones with limited online functions. Requiring a smartphone would define a different target with different marginal coverage.

All eight capability profiles are feasible: someone may use the internet on a computer or a borrowed phone, and an account holder may lack a phone, internet use, or both. We allow the distribution to assign mass to any of these profiles.

The release retrieved on 12 September 2026 contains all three marginals for 141 economies: 93 low- or middle-income economies and 48 high-income economies, using the source’s income classification. The official indicator-mapping workbook confirms this questionnaire coverage. More detailed connectivity activities cover fewer economies and are excluded.

Gender comparisons cover all 141 economies. Income comparisons cover 140, excluding Lesotho, for which both income-group rows are missing. Rural–urban comparisons cover 138: Bahrain and Singapore lack rural rows, and Eswatini lacks an urban row. The corresponding low- and middle-income samples contain 93, 92, and 92 economies. Every available subgroup row has all three marginals, and we impute no missing values. Cross-economy summaries give each economy equal weight and exclude regional aggregates; they describe the distribution of economy-level estimates.

8.2 Costs and available information

We set each incremental acquisition cost to one and the fixed delivery cost to $f = 1$ for each adult requiring an upgrade. Equations (16)–(17) give the cost bounds for completing all three capabilities; the intersection bounds describe the share already possessing them. Calculations use unrounded source proportions throughout.

Equal unit costs isolate the role of dependence under a fixed delivery technology. These assumptions express resource requirements in normalized units. Monetary budgets require programme-specific acquisition and delivery costs.

Bounds from published marginals measure the information lost by omitting joint outcomes. The microdata locate the weighted joint distribution within these bounds and show how much selected cross-tabulations narrow them (Section 10). All bounds condition on estimated moments and the assumed costs; Section 12 addresses sampling uncertainty.

9 Cross-economy and subgroup results

9.1 Uncertainty about joint inclusion

Marginal rates leave substantial uncertainty about joint inclusion (Table 1). Across 141 economies, the mean interval width is 22.37 percentage points and the median is 20.50 points. Among the 93 low- and middle-income economies, these figures rise to 29.32 and 29.74 points. In 39 of them, the lower bound is zero: the marginals are compatible with a population in which every adult lacks at least one capability.

This uncertainty is widespread. Fifty-three of the 93 low- and middle-income economies have intervals wider than 25 percentage points, and 21 have widths above 40 points. Tajikistan has the widest interval, at 54.52 points, followed by the Philippines at 50.18 and Nepal at 50.12. Width measures uncertainty about overlap. A scarce capability places a low ceiling on inclusion, while near-universal coverage raises its lower bound. Uncertainty can thus be greatest at intermediate coverage levels.

Consistent with this pattern, mean widths are 28.98 percentage points in low-income economies, 33.44 in lower-middle-income economies, 25.02 in upper-middle-income economies, and 8.90 in

Table 1: Uncertainty about possession of the complete capability bundle

Sample	Economies	Mean width	Median width	Zero lower bound
All economies	141	22.37	20.50	39
Low- and middle-income	93	29.32	29.74	39
High-income	48	8.90	6.72	0

Notes: Widths are percentage points of the inclusion interval. Each economy receives equal weight. At $f = 1$, dividing these widths by 100 gives the normalized cost-interval width. Income categories follow the source file. Source: Authors’ calculations using Global Findex 2025, 2024 reference wave.

high-income economies. Marginal shortfalls and uncertainty about their overlap are distinct features of the data.

9.2 Selected economies

In India, average access is 67.1%, and the share possessing all three capabilities can lie anywhere between 1.2% and 45.7% (Table 2). High account ownership leaves open whether account owners also own phones and use the internet. The corresponding cost interval is $[1.530, 1.975]$ normalized units per adult.

Indonesia has a similar average access rate, 69.0%, with account ownership as its smallest marginal. Its inclusion interval is $[7.1\%, 56.3\%]$ and its cost interval is $[1.365, 1.857]$. The overlap with India’s cost interval leaves the cost ranking unresolved.

Kenya’s average access exceeds 81%; its internet-use rate caps joint inclusion at 60.5%. Ethiopia combines a narrow inclusion interval, $[0, 13.8\%]$, with a high cost lower bound of 2.656. Its low internet use limits both inclusion and uncertainty about inclusion. China and Brazil have comparatively high inclusion lower bounds because their marginal rates sum to well above two. Their precise overlap remains unknown.

Account ownership sets the inclusion upper bound in 51 of the 93 low- and middle-income economies, and internet use in the other 42. Phone ownership is never the smallest marginal. These ceilings identify constraints on current inclusion; choosing an intervention also requires its costs, effects, take-up, and target population.

9.3 Group differences

The cost-gap bounds in (24) allow some group comparisons without knowing within-group dependence. A positive lower endpoint establishes a higher cost for the first group under every compatible pair of distributions; an interval containing zero leaves the sign unresolved. We examine gender, income, and residence separately. Comparisons involving intersections of these groups, such as poor rural women, require joint group data.

Income gaps are most often sign-identified (Table 3). The poorest group’s completion cost exceeds the richest group’s in all compatible distributions in 83 of 92 economies. The same holds for rural relative to urban adults in 62 of 92 economies, and for women relative to men in 42 of 93. Mongolia has a uniformly negative female-minus-male gap. The remaining 50 gender intervals contain zero, leaving the ordering unresolved.

Some gender gaps are large even at their lower bounds. Pakistan’s female-minus-male gap is $[1.269, 1.811]$ normalized units, India’s is $[0.282, 0.973]$, and Kenya’s is $[0.413, 0.754]$. Brazil’s interval, $[-0.011, 0.369]$, leaves the sign unresolved. Joint information can therefore matter for the magnitude of a gap, its sign, or both.

9.4 Sensitivity to delivery costs

Cost-interval widths rise linearly with the fixed delivery cost. At $f = 0, 0.25, 0.5, 1, 2$, their medians across the 93 low- and middle-income economies are 0, 0.0744, 0.1487, 0.2974 and 0.5948 normalized units per adult. Across all economies, the corresponding medians are 0, 0.0513, 0.1025, 0.2050 and 0.4101. With $f = 0$, completion cost is additive and point-identified even though joint inclusion remains only partially identified. These comparisons hold observed coverage fixed and vary only the assumed delivery cost.

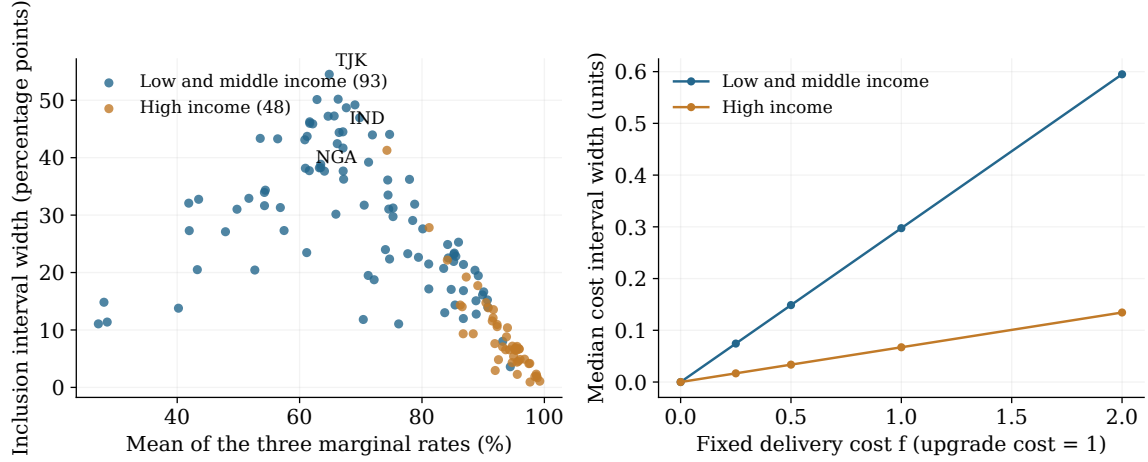


Figure 2: Access, ambiguity and cost sensitivity. Left: each point represents an economy in the 2024 reference wave. Right: median cost-interval width as the fixed delivery cost changes. The uncertainty concerns dependence conditional on published marginal estimates.

Table 2: Selected economies: marginal access, inclusion, and completion cost

Economy	Phone	Internet	Account	Average	Inclusion	Cost
Bangladesh	82.4	43.6	43.3	56.4	[0.0, 43.3]	[1.875, 2.307]
Brazil	92.1	87.5	86.4	88.7	[66.0, 86.4]	[0.476, 0.680]
China	96.6	86.2	89.4	90.8	[72.3, 86.2]	[0.415, 0.555]
Ethiopia	58.0	13.8	48.8	40.2	[0.0, 13.8]	[2.656, 2.794]
India	66.5	45.7	89.0	67.1	[1.2, 45.7]	[1.530, 1.975]
Indonesia	80.3	70.5	56.3	69.0	[7.1, 56.3]	[1.365, 1.857]
Kenya	92.7	60.5	90.1	81.1	[43.3, 60.5]	[0.962, 1.133]
Nigeria	83.8	37.7	63.3	61.6	[0.0, 37.7]	[1.775, 2.152]
Pakistan	62.9	35.7	27.3	42.0	[0.0, 27.3]	[2.468, 2.741]
South Africa	87.0	68.3	81.1	78.8	[36.4, 68.3]	[0.953, 1.272]

Notes: Marginals, their arithmetic average, and inclusion endpoints are percentages. Cost intervals are normalized units per adult with three unit incremental costs and $f = 1$. All endpoints use unrounded source inputs. The examples are illustrative; the complete 141-economy results accompany the paper. Source: Authors' calculations using Global Findex 2025.

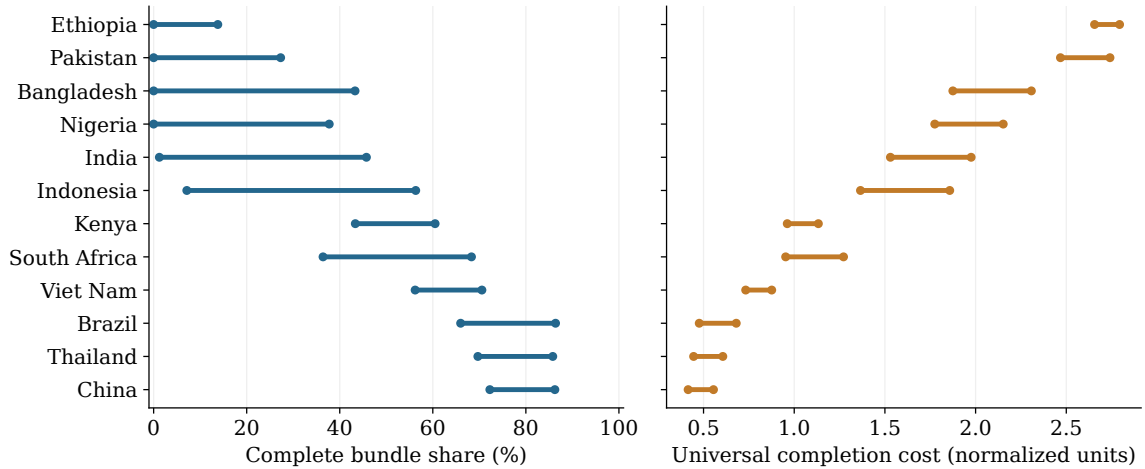


Figure 3: Sharp inclusion and completion-cost intervals in illustrative economies. The panels show joint inclusion (left) and normalized cost (right). Endpoints describe feasible overlap structures conditional on the three published marginal estimates.

Table 3: Sign identification of group cost gaps in low- and middle-income economies

First group versus reference	Pairs	Higher	Lower	Includes zero
Women versus men	93	42	1	50
Poorest 40% versus richest 60%	92	83	0	9
Rural versus urban	92	62	0	30

Notes: Higher denotes a strictly positive lower cost-gap endpoint; lower denotes a strictly negative upper endpoint. Counts describe sign identification conditional on published marginal estimates and normalized baseline costs; sampling inference is discussed in Section 12. Lesotho is omitted from the income comparison and Eswatini from the residence comparison because the required source rows are absent. Source: Authors' calculations using Global Findex 2025.

9.5 Coverage under a fixed budget

With a fixed budget, affordable inclusion depends on who needs which upgrades. Proposition 8, applied to the same marginals and costs, gives the smallest and largest affordable post-programme inclusion shares across compatible distributions. The lower endpoint is the adaptively guaranteed share in (22); the upper endpoint is best-case coverage. Targeting occurs after the programme learns capability profiles, treatment is divisible within states, and screening is funded separately. Budgets and costs are in normalized units per adult.

With a budget of one unit per adult, median guaranteed coverage among the 93 low- and middle-income economies is 61.22%, compared with median best-case coverage of 88.62%. Budgets of 0.5 and 1.5 yield guaranteed medians of 36.22% and 86.22%, and best-case medians of 71.07% and 100%. Each figure is a separate median of economy-specific endpoints. These coverage bounds condition on the assumed delivery technology; programme evaluation would require evidence on intervention effects.

Figure 4 plots the budget–coverage frontier for Brazil, India, and Nigeria. At zero budget, the endpoints equal the initial inclusion bounds. As the budget rises, each compatible population is served in order of increasing completion cost. Best-case coverage reaches one at the lower bound on universal-completion cost; guaranteed coverage reaches one at the upper bound. The frontier thus depends on the distribution of costs across adults, whereas universal completion depends only on their mean.

10 Observed overlap and the value of cross-tabulations

10.1 Joint inclusion in the microdata

Among the 93 low- and middle-income economies, the median share with all three capabilities is 37.79%. The median average access rate across the three dimensions is 70.41%.

The public-use Findex file contains 144,090 respondents in 140 economies (World Bank, 2025). It covers all 93 low- and middle-income economies in our baseline; the United Arab Emirates is the only baseline economy absent. We retain the baseline income classification and the official indicator definitions. Estimates within each economy use final respondent weights, while summaries across economies give each economy equal weight. Appendix D documents coding, missing demographic labels and reconciliation with the published aggregates.

For each economy, we estimate the eight-state distribution by

$$\hat{\mu}_x = \frac{\sum_i w_i \mathbf{1}\{X_i = x\}}{\sum_i w_i}, \quad \hat{s} = \hat{\mu}_{111}, \quad \hat{T} = \sum_{j=1}^3 (1 - \hat{p}_j) + 1 - \hat{s}. \quad (29)$$

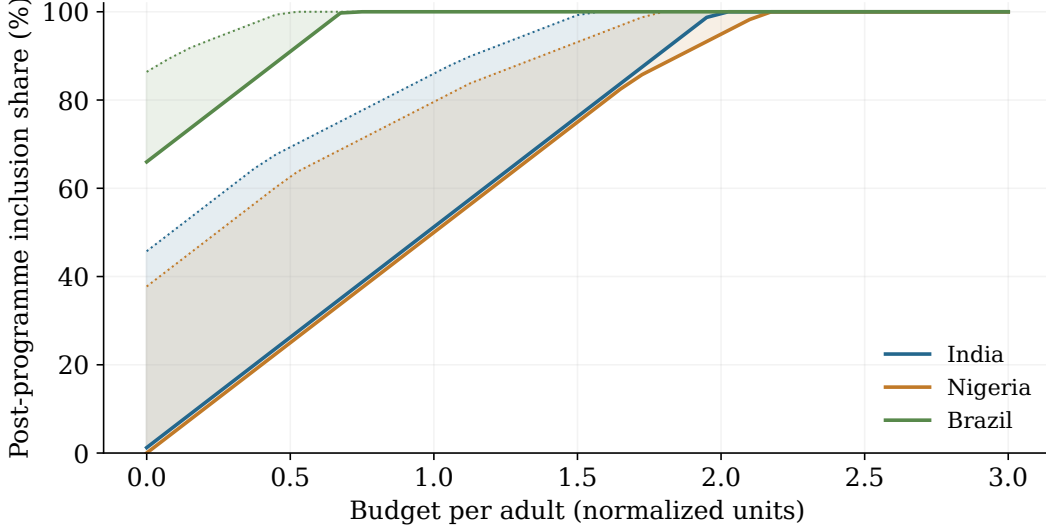


Figure 4: Affordable inclusion under alternative dependence structures. Solid lines show adaptively guaranteed coverage and dotted lines best-case coverage; shaded regions are the sharp intervals. Calculations use published economy marginals and assumed normalized costs.

We compare bounds based on the three marginals, bounds that also use pairwise overlaps, and estimates from the full joint distribution. All moments come from the same sample, coding and weights, so the comparison isolates the value of additional information. The intervals describe identification uncertainty conditional on estimated moments; Section 12 discusses sampling uncertainty.

The estimated joint share exceeds the product of the marginals in all 140 economies. The median excess is 6.08 percentage points overall and 7.87 points in the low- and middle-income sample. Thus, at $f = 1$, assuming independence overstates the estimated completion cost by a median of 0.0608 and 0.0787 normalized units per adult, respectively.

10.2 Pairwise information and the remaining third-order ambiguity

Write $q_{jk} = \Pr(X_j = X_k = 1)$. With all three marginals and all three pairwise moments known, all eight cells of the three-dimensional binary distribution can be parameterized by $s = \Pr(X = 111)$. Nonnegativity of its eight cells gives the sharp bounds

$$s_{\text{pair}}^- = \max\{0, q_{12} + q_{13} - p_1, q_{12} + q_{23} - p_2, q_{13} + q_{23} - p_3\}, \quad (30)$$

$$s_{\text{pair}}^+ = \min\{q_{12}, q_{13}, q_{23}, 1 - p_1 - p_2 - p_3 + q_{12} + q_{13} + q_{23}\}. \quad (31)$$

Indeed, the cells with exactly two capabilities are $q_{jk} - s$, those with exactly one are $p_j - q_{jk} - q_{j\ell} + s$, and the empty cell is $1 - \sum_j p_j + \sum_{j < k} q_{jk} - s$. Every value between the displayed endpoints makes these cells nonnegative and reproduces all six moments. Pairwise releases can therefore narrow the interval without determining the three-way overlap. Cost bounds follow by substituting s_{pair}^+ and s_{pair}^- , respectively, into $D + 1 - s$, where $D = \sum_{j=1}^3 (1 - p_j)$.

10.3 How much information do actual pairwise releases add?

Releasing all three pairwise overlaps reduces the median inclusion interval width from 29.74 to 2.11 percentage points (Table 4). The median proportional reduction, calculated within each economy, is 91.33%. The remaining width exceeds five percentage points in only five of the 93

economies. At $f = 1$, the cost interval narrows in the same proportion because its width equals the inclusion width in probability units.

The internet–account overlap is the most informative single-pair release in 85 of the 93 economies. Phone–internet and phone–account yield the narrowest interval in four economies each, with no ties at the numerical tolerance. This ranking uses observed overlaps for the target and technology considered here. The direct three-way share closes the remaining interval using one number. Comparing equal-precision releases therefore favours this targeted statistic when the sole objective is baseline inclusion or cost.

10.4 Selecting a pair from marginal rates

The prospective rule in Proposition 9 selects the most informative observed pair in all 93 low- and middle-income economies and all 140 economies with microdata. Its median remaining width is 11.93 percentage points in the main sample, a within-economy median reduction of 59.31%. Ten economies have multiple minimax pairs; choosing the pair that omits a largest marginal selects the unique realized best pair in each. This agreement is a national-sample finding; the appendix gives a distribution where the two choices differ.

Marginal ordering accounts for the 85/93 pattern: phone ownership is the highest marginal in exactly those 85 economies. Internet use and account ownership are highest in four economies each. Among the 85, the internet–account width equals the phone deficit in 72; in the remaining 13, it equals the observed internet–account overlap. Both branches follow from (26). The ranking therefore reflects the high phone coverage together with the observed dependence.

India illustrates the rule’s economy-specific choice. Estimated joint inclusion is 37.99% and completion cost is 1.608 normalized units per adult. The marginal-only inclusion interval has width 44.50 points. A phone–internet overlap reduces that width to 10.98 points, while the internet–account overlap leaves 33.53 points. All three pairs reduce the width to 3.27 points. In Nigeria, by contrast, the internet–account overlap narrows the interval from 37.74 to 16.23 points; all three pairs leave only 1.59 points. Nigeria’s weighted joint share is 33.42%, compared with 20.00% under independence. Delivery costs are held fixed across these comparisons; the differences come from the margins and overlaps. Account ownership is India’s largest marginal, so the rule selects phone–internet using the published rates alone.

Table 4: Actual information refinement in 93 low- and middle-income economies

Information released	Median width	Mean width	Median reduction (%)
Three marginals only	29.74	29.32	0.00
+ Phone–internet overlap	22.91	23.13	17.93
+ Phone–account overlap	21.19	23.43	20.04
+ Internet–account overlap	12.29	12.85	58.75
+ Pair selected from marginals	11.93	12.44	59.31
+ All three pairwise overlaps	2.11	2.39	91.33
+ Direct three-way share	0.00	0.00	100.00

Notes: Widths are percentage points of the sharp inclusion interval. Each single-pair row adds only that pair to the three marginals. The marginal-based rule omits a dimension with the largest marginal, as in Proposition 9. The direct three-way share is one additional statistic for the baseline target. Reduction is $100(1 - W_{\text{release}}/W_{\text{marginals}})$, calculated within each economy before taking the median. Each economy receives equal weight. All moments use the same final-weighted sample. Source: Authors’ calculations using Global Findex public-use microdata.

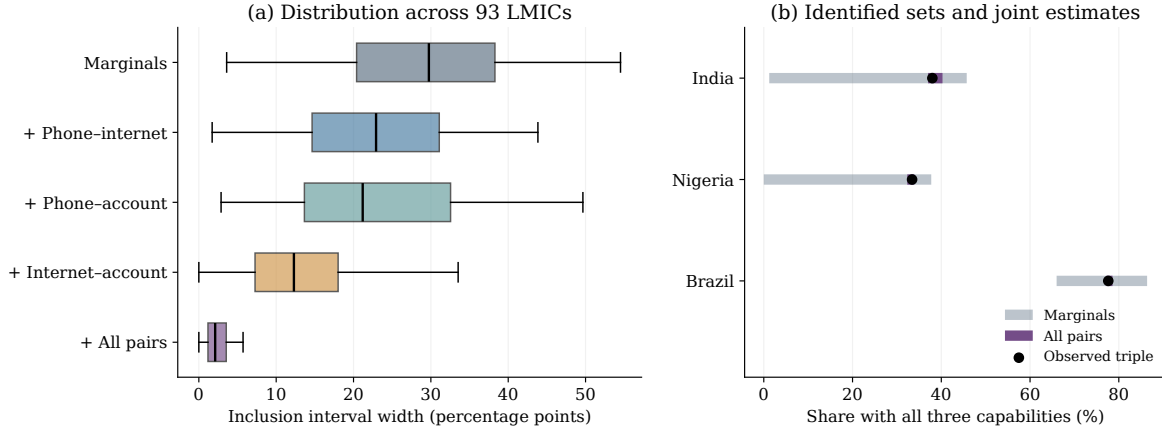


Figure 5: Observed joint inclusion and actual information refinement. All calculations use the same weighted respondent sample within each economy. Left: boxes show the interquartile range and median across economies; whiskers use the 1.5-interquartile-range rule and outliers are hidden. Single-pair additions are alternative releases. Points show weighted estimates of joint inclusion.

10.5 Budget feasibility and group comparisons

Additional information can settle a decision even when some interval width remains. For a universal-completion budget B , let $[L, U]$ be the cost interval under the available release. Completion is affordable under every compatible distribution if $U \leq B$, unaffordable under every distribution if $L > B$, and unresolved if $L \leq B < U$. We compare releases over the budget grid $B \in \{0.5, 1, 1.5, 2\}$, using the baseline normalized costs.

At $B = 2$, the marginals leave affordability unresolved in 23 of 93 economies. The prospective single-pair rule reduces this to five, and all three pairs resolve affordability in all 93 economies at this budget. At $B = 1$ and $B = 1.5$, all pairs leave one economy unresolved; the direct three-way share resolves it (Table 5). These are classifications within the assumed technology, conditional on survey estimates.

Table 5: Unresolved comparisons under alternative information releases

Comparison	Marginals	One pair	All pairs	Three-way
Universal-completion budget, 93 economies				
$B = 0.5$	6	2	0	0
$B = 1$	5	3	1	0
$B = 1.5$	13	5	1	0
$B = 2$	23	5	0	0
Sign of the per-person completion-cost gap				
Women–men (93)	50	13	3	0
Poorest 40%–richest 60% (92)	9	0	0	0
Rural–urban (92)	30	10	1	0

Notes: Counts across low- and middle-income economies. One pair is selected from marginal rates by omitting a largest marginal. Group comparisons select and release an overlap separately for each group, requiring two additional numbers per comparison. All pairs and three-way shares are likewise group-specific. A gap is unresolved when its interval contains zero. Budgets are normalized units per adult. All calculations condition on the same weighted moments and the baseline cost technology. Source: Authors' calculations using Global Findex public-use microdata.

Selecting one pair separately within each group resolves all nine income comparisons left ambiguous by marginals, 37 of 50 gender comparisons, and 20 of 30 rural–urban comparisons. All pairs leave three gender comparisons and one residence comparison unresolved. Using the full

joint distribution, estimated cost is higher for women in 82 of 93 economies, for the poorest 40% in 91 of 92, and for rural adults in 89 of 92. These are descriptive signs; population comparisons require the inference in Section 12.

The pair chosen from marginals yields a wider interval than the pair chosen after observing overlaps in five of 277 subgroup comparisons. The largest difference is 0.0235 normalized cost units per adult, for income groups in Honduras. Both rules give the same sign classifications throughout the sample: minimizing interval width and resolving a group ranking are distinct objectives.

10.6 What the capability bundle measures

Requiring the main phone to be a smartphone lowers the median joint share from 37.79% to 36.07% across the same 93 low- and middle-income economies (Table 6). The median within-economy decline is 2.90 percentage points. The median identification width rises to 33.60 points under marginals and 2.73 points with all pairs. Both coverage and its information requirement therefore depend on the capability standard.

Table 6: Alternative device requirements for the capability bundle

Device requirement	Joint share (%)	Marginal width	All-pair width
Any personal phone	37.79	29.74	2.11
Main phone is a smartphone	36.07	33.60	2.73

Notes: Each entry is a separate median across the same 93 economies. Widths are percentage points. Both bundles also require internet use and account ownership. The stricter device measure requires phone ownership and an affirmative response that the main phone used is a smartphone; other adults remain in the denominator. Both definitions use the same 97,923 respondents and weights. Source: Authors' calculations using Global Findex public-use microdata.

We also compare the baseline bundle with the official indicator of making or receiving a digital payment in the past year. This module covers 92 of the 93 economies, with Russia absent. Among adults with the full bundle, the economy-level median payment rate is 91.54%; among adults with an incomplete bundle, it is 20.17%. The within-economy difference is positive in all 92, with a median of 63.66 percentage points. Adults who have the full bundle and report no digital payment account for a median 4.42% of all adults.

These comparisons relate the bundle to financial behaviour. The payment indicator includes transfers and card transactions, and its definition overlaps with account ownership. Independent assessments of sustained use, autonomy and safety require additional evidence.

11 Alternative technologies and capability standards

The cost of overlapping deficits depends on the delivery technology. In the baseline, concentrating shortfalls reduces the number of enrolments. Adults with several shortfalls may also require more intensive assistance. With $N(x) = \sum_{j=1}^3 (1 - x_j)$, consider

$$h_\gamma(x) = N(x) + \gamma \frac{N(x)(N(x) - 1)}{2}, \quad \gamma \in \{0.5, 1\}. \quad (32)$$

This specification replaces the setup fee with an additional charge for each pair of missing capabilities. As in the baseline, the coefficients are assumed normalized costs.

To represent (32) on the capability graph, give each edge that adds a missing capability the cost $c_j(x) = 1 + \gamma[N(x) - 1]$. Every completion path reduces the missing count from $N(x)$ to

zero and therefore sums to $h_\gamma(x)$. Edge costs decrease as other capabilities improve, satisfying the sufficient monotone-technology condition. Overlapping deficits now raise costs: nested deficit events maximize all pairwise overlaps and attain the upper bound.

The information requirement also changes. Since

$$\mathbb{E}h_\gamma(X) = \sum_j (1 - p_j) + \gamma \sum_{i < j} \Pr(X_i = 0, X_j = 0), \quad (33)$$

all pairwise tables identify this cost. The baseline cost also depends on the three-way complete-bundle probability, which can remain partially identified with all pairwise tables. The technology determines which statistics are needed.

Table 7: Technology and target sensitivity in 93 low- and middle-income economies

Scenario	Median lower	Median upper	Median width
Three capabilities, setup fee 1	1.455	1.776	0.297
Three capabilities, convex cost $\gamma = 0.5$	0.888	1.151	0.222
Three capabilities, convex cost $\gamma = 1$	0.888	1.421	0.443
Phone and internet, setup fee 1	0.702	0.879	0.126

Notes: Normalized units per adult; each economy receives equal weight. Each column reports its own median across economies; the median width is computed from economy-specific widths. Convex-cost scenarios replace the setup fee. Bounds condition on the same published marginal estimates. Source: Authors' calculations using Global Findex 2025, 2024 reference wave.

Table 7 reports results for the 93 low- and middle-income economies from calculations for all 141 economies on the eight-state cube. The median width is 0.222 when $\gamma = 0.5$ and 0.443 when $\gamma = 1$, compared with 0.297 in the baseline. At $\gamma = 1$, India's interval is $[0.988, 1.542]$, Nigeria's is $[1.305, 1.844]$, and Brazil's is $[0.340, 0.624]$. Dependence uncertainty remains substantial when overlapping shortfalls raise expenditure. Differences in cost levels across specifications also reflect the assumed charging rules.

Finally, we relax the target to phone ownership and internet use, retaining unit incremental costs and a setup fee of one. Its cost is $h_{PI}(x) = (1 - x_1) + (1 - x_2) + \mathbf{1}\{(x_1, x_2) \neq (1, 1)\}$; account ownership is optional. Both cost endpoints weakly fall in every economy, and the median width falls to 0.126 in the low- and middle-income sample. Pakistan's width rises from 0.273 to 0.357 when the target no longer requires account ownership, its least prevalent capability. A weaker target reduces the required resources and can leave more uncertainty about their amount.

12 Sampling uncertainty

The reported bounds hold survey estimates fixed and vary the unobserved joint distribution. Statistical inference on population gaps also requires accounting for sampling uncertainty. The public microdata provide final weights; design-based inference additionally requires stratum and primary sampling-unit identifiers or replicate weights, which are unavailable in the public release. We therefore report descriptive estimates and leave standard errors and confidence intervals for work with the full survey design. Adjustments for nonresponse, measurement error, and differences in survey coverage or interview mode also fall outside the calculation.

12.1 Confidence regions for the identified set

A simultaneous confidence region for the moments provides a direct route to inference. Let $\mathcal{P}_n$ cover the population marginal vector p_0 , and let $\mathcal{M}$ be the feasible moment space, equal to $[0, 1]^d$ in the unrestricted model. Suppose

$$\Pr\{p_0 \in \mathcal{P}_n\} \geq 1 - \alpha. \quad (34)$$

For nonempty $\mathcal{P}_n \cap \mathcal{M}$, define

$$\mathcal{C}_n = \left[\inf_{p \in \mathcal{P}_n \cap \mathcal{M}} \underline{T}(p), \sup_{p \in \mathcal{P}_n \cap \mathcal{M}} \bar{T}(p) \right]. \quad (35)$$

Whenever $p_0 \in \mathcal{P}_n$, this interval contains the entire population identified set $[\underline{T}(p_0), \bar{T}(p_0)]$. Equation (34) thus gives coverage of that set; an asymptotically valid moment region gives the corresponding asymptotic result. The same argument applies to joint moments and group differences if the region covers all moments entering the calculation simultaneously. It conditions on the maintained support and technology. Covering the identified set differs from covering a particular parameter within it (Imbens and Manski, 2004).

The confidence region must account for survey stratification, clustering, weighting and covariance across indicators (Lumley, 2010). Design-valid marginal intervals with a familywise adjustment offer one conservative construction. The coverage family must include every economy or subgroup for which a simultaneous claim is intended. Projection requires a nonempty $\mathcal{P}_n \cap \mathcal{M}$; an empty intersection calls for examination of the data and maintained restrictions.

Projection accommodates the kinks in bound endpoints produced by maxima and minima. Conventional bootstrap inference may fail at these kinks (Fang and Santos, 2019). Direct endpoint inference can be less conservative with a procedure valid for such nonsmooth functions.

12.2 What the respondent data establish

The empirical joint distribution and its marginals use the same records and weights. Its inclusion and cost estimates must therefore lie within the corresponding bounds. This provides an algebraic check. Validating the intervention technology requires evidence on programme delivery; confidence statements about the population require valid sampling inference.

The substantive comparison measures how much joint information narrows the bounds. It requires common questionnaire universes, coding and denominators so that all moments describe the same empirical distribution. Complete-case restrictions define a common sample with all required data; extrapolation beyond that sample requires assumptions about missingness. Appendix D documents our construction.

13 Conclusion

Marginal coverage rates are enough for some policy questions and inadequate for others. They identify the cost of giving everyone the full bundle when upgrade costs are constant and additive. A fee for each recipient changes the calculation: the programme must know how many distinct people need assistance. More generally, the interactions in minimum completion costs determine which joint moments matter.

The Findex application quantifies this information loss and its consequences. Across 93 low- and middle-income economies, marginal rates leave a median inclusion interval of 29.74 percentage points. Choosing one pair by marginal ordering reduces it to 11.93 points; all pairs

reduce it to 2.11. The prospective rule selects the most informative observed pair in every national sample. These releases resolve budget and group comparisons, with some subgroup intervals showing a difference between prospective and retrospective selection.

Release design starts with the intended use. A direct three-way share identifies the baseline target and cost with one additional statistic. The single-pair rule solves a specified equal-cost menu; actual reporting costs, privacy mechanisms and multiple policy objectives would define other design problems. The device and payment comparisons show how the bundle relates to selected access conditions and financial behaviour. Independent evidence on autonomy, safety and sustained use would support a broader capability interpretation.

Our normalized scenarios isolate the role of dependence. Monetary budgets require delivery costs, take-up, effectiveness and capacity evidence for a specified programme. Design-based confidence statements require additional survey information. The practical task is to define the policy standard and technology, then collect the joint statistics and implementation evidence needed to evaluate them.

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A Proofs and constructive arguments

A.1 Transport reduction

Every unit sent from x to A costs at least $h_A(x)$, so every feasible transport costs at least $\sum_x \mu_x h_A(x)$. Finiteness and reachability ensure a minimizing destination y_x exists. Sending all mass at each x to y_x attains the bound because choices at one source do not restrict choices at another.

A.2 Proof of Theorem 2

Let B have column $(1, x^\top)^\top$ at state x , and let $m = (1, p^\top)^\top$. A product Bernoulli distribution with probabilities p belongs to $\mathcal{F}(p)$, including at boundary values. Thus $\{\mu \geq 0 : B\mu = m\}$ is a nonempty, closed subset of the probability simplex. Compactness gives both extrema, and mixtures of their optimizers attain every value between them.

The dual of $\min\{h^\top \mu : B\mu = m, \mu \geq 0\}$ is $\max\{\theta^\top m : B^\top \theta \leq h\}$. Feasibility and boundedness imply strong linear-programming duality and dual attainment. Reversing the objective gives the upper formula. An extreme distribution cannot have more than $\text{rank } B$ positive entries: linear dependence of the corresponding columns would give a nonzero v with $Bv = 0$, allowing small feasible perturbations in both directions. Since linear objectives attain extrema at extreme points, an optimizer can be chosen with at most $d + 1$ positive entries.

The lower value is a supremum of affine functions of p , hence convex; the upper value is an infimum, hence concave. Complementary slackness restricts positive mass to states where the affine bound equals h .

A.3 Proof of Theorem 3 and its support extension

If $h = B^\top \theta$, then $h^\top \mu = \theta^\top m$ for every feasible law. Conversely, at interior p the product Bernoulli law μ^0 is strictly positive. For any $v \in \ker B$, both $\mu^0 \pm \varepsilon v$ are feasible for sufficiently small $\varepsilon > 0$. A constant objective therefore requires $h^\top v = 0$ for every $v \in \ker B$. Hence

$$h \in (\ker B)^\perp = \text{range}(B^\top),$$

which is precisely the space of affine functions on the cube.

At a boundary vector, coordinates with mean zero or one are fixed almost surely. The product law is strictly positive on the remaining face, where the same argument applies.

For general observed functions, choose for each active state a feasible law assigning it positive mass. Averaging these laws gives a feasible law positive throughout $S^*(m)$. Apply the same null-space argument on this support; inactive states never affect the expectation.

The products $\prod_{j \in J} x_j$ form a basis of functions on the cube: evaluation at subset indicator vectors gives a triangular inclusion matrix. If all moments through order r are observed and a fully supported law is feasible, the span result identifies exactly the functions with no terms above degree r .

A.4 Proof of Theorem 4

Let M be a matrix whose rows are the observed functions, including a constant, and let $V = \text{range}(M^\top)$. Maximizing the width over feasible moment vectors is equivalent to

$$\max \left\{ h^\top (\mu - \nu) : \mu, \nu \in \Delta(\mathcal{S}), M\mu = M\nu \right\}. \quad (36)$$

The set of possible differences $z = \mu - \nu$ equals

$$\{z : Mz = 0, \|z\|_1 \leq 2\}.$$

The forward inclusion follows from the moment restrictions and the triangle inequality. Conversely, the constant moment implies $\sum_x z_x = 0$, so the positive and negative parts have equal mass $t = \|z\|_1/2 \leq 1$. For any probability vector r , let $\mu = z_+ + (1-t)r$ and $\nu = z_- + (1-t)r$. These are probability distributions with the same moments and difference z .

The dual of the uniform-approximation linear programme gives

$$\inf_{v \in V} \|h - v\|_\infty = \max \{ h^\top z : Mz = 0, \|z\|_1 \leq 1 \}.$$

Scaling the constraint by two and using (36) proves equality. The distribution-pair feasible set is compact, and the approximation subspace is closed, so both optima are attained. For any equal-moment pair and $v \in V$, the expectation difference for h is at most $2\|h - v\|_\infty$. Minimizing over v gives the pointwise bound.

A.5 Proof of Proposition 5

Under Assumption 1, $-h$ is increasing. It is a constant plus a nonnegative linear combination of indicators of its upper level sets, each upward closed. Dominance on those sets implies $\mathbb{E}_{\mu'}[-h] \geq \mathbb{E}_\mu[-h]$.

Start with a minimizing law at p . Raise each mean to p'_j by randomly changing enough zeros to ones in coordinate j , leaving other coordinates unchanged. The resulting law at p' dominates the original and has no higher cost, so $\underline{T}(p') \leq \underline{T}(p)$. For the upper endpoint, start with a maximizing law at p' and randomly change ones to zeros to obtain marginals p . Its cost is no lower, proving $\overline{T}(p') \leq \overline{T}(p)$. Both constructions use unrestricted support.

A.6 Proof of Proposition 6

Take a law with marginals p . To raise coordinate j to $q_j > p_j$, flip each zero to one with probability $(q_j - p_j)/(1 - p_j)$. To lower it, flip each one to zero with probability $(p_j - q_j)/p_j$. The denominators are nonzero whenever a change is needed. Other means are unaffected, the probability of a flip is $|p_j - q_j|$, and a flip changes h by at most L_j . Sequential changes produce a law at q whose expected cost differs by at most $\sum_j L_j |p_j - q_j|$.

Apply this construction to each endpoint optimizer and then interchange p and q to obtain both inequalities. Similarly, $\|h - \tilde{h}\|_\infty \leq \eta$ changes either endpoint by at most η . On an unchanged graph, perturbing each edge cost by at most ϵ changes a minimum completion cost by at most $d\epsilon$, since a strictly increasing path has at most d edges.

A.7 Information refinement and group decomposition

Additional valid moments shrink the feasible set, raising the minimum and lowering the maximum. The span result gives the conditions for point identification. With no restrictions across groups, the feasible class is a Cartesian product: choosing each group's optimizer attains the extreme sum. Combining one group's minimum with the other's maximum attains the lower gap endpoint; reversing the choices attains the upper endpoint. Mixtures fill the intervals.

For a common cost function across groups, pooled marginals satisfy $p = \sum_g w_g p_g$. Convexity and concavity give

$$\underline{T}(p) \leq \sum_g w_g \underline{T}(p_g) \leq \sum_g w_g \overline{T}(p_g) \leq \overline{T}(p).$$

Thus consistent marginal tables for disjoint groups weakly narrow the pooled interval.

A.8 Proof of Corollary 7

Let E_j be the event that capability j is absent, with probability D_j . The union satisfies

$$\max_j D_j \leq \Pr\left(\bigcup_j E_j\right) \leq \min\left\{1, \sum_j D_j\right\}.$$

Nested events of sizes D_j attain the lower endpoint. For the upper endpoint, place intervals of lengths D_j consecutively on the line and project them onto a unit circle. Each retains measure $D_j \leq 1$. The intervals are disjoint up to endpoints if their total length is at most one; otherwise they cover the circle. Under the uniform law their union therefore has measure $\min\{1, \sum_j D_j\}$. The resulting binary profile has the required marginals.

Both constructions have additive cost K and incur the fee exactly on the union, proving (16)–(18) and endpoint attainment. Complete inclusion is the complement of this union.

A.9 Proof of Proposition 8

For $t \geq 0$, the Lagrangian for the coverage constraint is

$$t\rho + \sum_x (h(x) - t)q_x.$$

Minimizing over $0 \leq q_x \leq \mu_x$ gives $t\rho - \sum_x \mu_x(t - h(x))_+$. Feasibility for $\rho \leq 1$ and boundedness give strong duality. The dual objective is continuous, concave and affine between consecutive costs. Above the largest cost its slope is $\rho - 1 \leq 0$, so a maximizer lies in H .

Maximization over $\mu \in \mathcal{F}(p)$ commutes with maximization over finite t . For each t , maximizing the negative expectation minimizes $\mathbb{E}(t - h)_+$, giving (21). The lower bound combines the two minimizations into one programme. As a maximum of finitely many affine functions, $C_\rho(\mu)$ is continuous. Its image over the connected, compact moment class is an attained interval, proving sharpness.

For a known law, $C_\rho(\mu) \leq B$ is equivalent to $R(B; \mu) \geq \rho$. It holds for every feasible law exactly when $\overline{C}_\rho(p) \leq B$. Taking the largest such ρ proves (22), which can be computed as

$$R_{\text{ad}}(B; p) = \min \left\{ 1, \min_{t \in H: t > 0} \frac{B + \min_{\mu \in \mathcal{F}(p)} \mathbb{E}_\mu(t - h(X))_+}{t} \right\}. \quad (37)$$

If all costs vanish, the inner set is empty and coverage is one. Best-case affordable coverage is a single joint programme maximizing $\sum_x q_x$ over $\mu \in \mathcal{F}(p)$, $0 \leq q \leq \mu$ and $h^\top q \leq B$.

B Why adaptive and precommitted delivery can differ

A policy may fix the treatment fraction $a_x \in [0, 1]$ for each state before learning the eligible pool's composition. Its robust budget requirement is

$$B_\rho^{\text{fixed}}(p) = \min_{0 \leq a \leq 1} \left\{ \max_{\mu \in \mathcal{F}(p)} \sum_x h(x) a_x \mu_x : \min_{\mu \in \mathcal{F}(p)} \sum_x a_x \mu_x \geq \rho \right\}. \quad (38)$$

Every feasible fixed policy supplies a feasible selection for each population, so $B_\rho^{\text{fixed}}(p) \geq \bar{C}_\rho(p)$. Both regimes observe individual states at delivery; only the adaptive policy adjusts its fractions to the population's composition.

The inequality can be strict even under decreasing completion costs. On the two-dimensional cube, use the state ordering (00, 10, 01, 11), marginal probabilities (3/4, 1/2), costs (2, 1, 1/4, 0) and coverage requirement $\rho = 3/4$. Compatible laws are

$$\mu(t) = (t - 1/4, 3/4 - t, 1/2 - t, t), \quad 1/4 \leq t \leq 1/2.$$

At $t = 1/4$ least-cost coverage costs 5/16; at $t = 1/2$ it costs 1/4. Convexity of the selection value in μ implies that the maximum on this segment is 5/16.

For a fixed policy set $a_{11} = 1$ without loss. Endpoint coverage restrictions imply $2a_{10} + a_{01} \geq 2$ and $a_{00} + a_{10} \geq 1$. Consequently $a_{10} \geq 1/2$ and worst-case cost is at least

$$\max\{(3/8)a_{10} + 1/8, 1/2 - (1/4)a_{10}\}.$$

This expression is minimized at $a_{10} = 3/5$, with value 7/20. Fractions $(a_{00}, a_{10}, a_{01}, a_{11}) = (2/5, 3/5, 4/5, 1)$ attain that bound. Hence

$$\bar{C}_{3/4}(p) = 5/16 < 7/20 = B_{3/4}^{\text{fixed}}(p).$$

The monotone single-coordinate graph with costs $c(00, 10) = 1$, $c(10, 11) = 1$, $c(00, 01) = 7/4$ and $c(01, 11) = 1/4$ generates these completion costs. The gap therefore arises from the timing of allocation decisions.

Dualizing the two moment problems converts (38) to a linear programme. With affine cost majorant (α_c, β_c) and coverage minorant (α_r, β_r) , minimize b subject to

$$\begin{aligned} 0 \leq a_x \leq 1, \quad \alpha_c + \beta_c^\top x &\geq h(x) a_x \quad \forall x, \\ \alpha_c + \beta_c^\top p &\leq b, \\ \alpha_r + \beta_r^\top x &\leq a_x \quad \forall x, \\ \alpha_r + \beta_r^\top p &\geq \rho. \end{aligned}$$

Other commitments, such as a common package for all recipients, change the feasible policy class.

C Proof and scope of the release rule

C.1 Proof of Proposition 9

Proof. The (X_i, X_j) cell probabilities are q , $p_i - q$, $p_j - q$ and $1 - p_i - p_j + q$; all are nonnegative exactly when $q \in [\ell_{ij}, u_{ij}]$. Let $B = \{X_i = X_j = 1\}$, with probability q . The intersection of

B and $\{X_k = 1\}$ has the displayed lower and upper bounds. To attain any s between them, allocate mass s with $X_k = 1$ inside B and mass $p_k - s$ outside B . Within each part, assign X_k independently of the remaining variation in (X_i, X_j) , with probabilities s/q and $(p_k - s)/(1 - q)$, respectively. Parts of zero mass need no assignment. The bounds ensure that these probabilities lie in $[0, 1]$, preserving all three marginals and the pairwise overlap. Subtracting the endpoints gives the width formula.

Set $a = \min\{p_k, 1 - p_k\} \leq 1/2$. Maximizing $\min\{q, 1 - q, a\}$ over $[\ell_{ij}, u_{ij}]$ gives $\min\{a, u_{ij}, 1 - \ell_{ij}\}$: a width of at least t , for $0 \leq t \leq a$, is possible exactly when the intervals $[\ell_{ij}, u_{ij}]$ and $[t, 1 - t]$ intersect. Substitution yields (27). For every pair, $1 - p_k \geq 1 - p_{\max}$ and $2 - p_i - p_j \geq 1 - p_{\max}$, since $p_i + p_j \leq 1 + p_{\max}$. Thus $R_{ij}(p) \geq \min\{p_{\min}, 1 - p_{\max}\}$. Taking $p_k = p_{\max}$ attains this bound. $\square$

C.2 Prospective and realized release choices

The minimax rule controls the worst width before the overlap is observed. Its realized width can exceed that of another pair. For example, assign masses 0.4, 0.3, 0.2 and 0.1 to states 001, 011, 110 and 111. The marginals are (0.3, 0.6, 0.8). The rule selects pair 12, whose worst width is 0.2, compared with 0.3 for each alternative. The observed overlaps are $q_{12} = 0.3$, $q_{13} = 0.1$ and $q_{23} = 0.4$. Their realized widths are 0.2, 0.1 and 0.3, respectively. This separates the prospective criterion from the ranking after observing the overlaps.

D Data construction and replication

D.1 Sources and sample

The replication package contains the public aggregate CSV, indicator glossary, indicator-to-economy mapping, derived results, analysis scripts and figures. The aggregate data were retrieved from the World Bank’s [Global Findex download page](#) on 12 September 2026. Download URLs, timestamps and SHA-256 hashes are recorded in `data/source_manifest.json`.

The microdata release (catalogue reference `WLD_2024_FININDEX_v02_M`) was obtained from the Microdata Library on 8 October 2026 under its access terms. It contains 144,090 records and 199 variables. Its version and hash are recorded in `data/microdata_source_manifest.json`. Licensed respondent records are excluded from the distributable package; users must obtain their own copy to reproduce Section 10.

Table 8: Sample construction

Selection	Rows or economies
Original public CSV, all reference waves	8,577 rows
Rows with reference year 2024	1,982 rows
Economy-by-demographic-group rows in 2024	1,826 rows
All-adult economy rows	141 economies
Low- and middle-income all-adult rows	93 economies
High-income all-adult rows	48 economies
Missing selected marginals within available 2024 rows	0 cells
Public-use respondent file	144,090 records
Economies in respondent benchmark	140 economies
Matched economy-by-domain cells	1,813 cells
Matched marginal indicators	5,439 values

In the aggregate file, `con1`, `internet` and `account_t_d` are proportions. We select the 2024 wave, distinguish economies from regional and income aggregates using the source’s region field, retain its income classification, and check uniqueness of economy–year–group keys. Population metadata on subgroup rows give national adult counts; subgroup calculations exclude these counts from their weights.

D.2 Respondent coding and reconciliation

Phone ownership is `con1==1`; no, don’t know and refusal remain in the denominator and are coded zero, as in the official indicator. Internet use is `internet_use`, checked against a yes to either `con24` or `con25`. Account ownership is the combined indicator `account`. These three variables and final weights are observed for every record, and all weights are positive. We retain structural skips in `con25`: respondents who used the internet in the preceding seven days need not answer the three-month follow-up. The combined account indicator likewise retains respondents whose mobile-money component is not separately recorded.

The 5,439 weighted marginal estimates across 1,813 economy–domain cells match the aggregate release to within 10^{-15} in probability units. The cells cover all adults and gender, income, residence, age, education and labour-force domains. The 13 unmatched cells all belong to the United Arab Emirates, which is absent from the respondent file. Missing income, residence, education and labour-force labels affect 1,020, 2,526, 588 and 4,020 records, respectively. Each exclusion applies only to the corresponding domain analysis; all-adult estimates retain these records. Final weights, `wgt`, apply within economies and domains. A global adult estimate would require an additional scheme for weighting economies.

D.3 Computation checks

`analysis/analyze_findex.py` computes the baseline bounds, subgroup gaps and summary statistics. `analysis/verify_and_plot.py` checks the closed forms against independent primal and dual linear programmes for all 1,982 available 2024 rows, including official aggregate rows outside the country sample. The maximum endpoint discrepancy is 8.9×10^{-16} and the maximum moment residual is below 2.3×10^{-16} .

The same script checks the affine-approximation identity on 40 random state-cost vectors, with discrepancies below 5×10^{-16} . For 40 random marginal–coverage cases, the finite dual formula for partial coverage agrees with enumeration of the Fréchet polytope’s basic feasible distributions to within 10^{-8} . Seeds and results are saved with the code. These are numerical checks, separate from the survey evidence.

`analysis/technology_robustness.py` computes the alternative cost and target scenarios on the eight-state cube. The number of states grows exponentially with dimension, so larger applications may require additional structure. `analysis/analyze_microdata.py` reproduces the respondent estimates and reconciliation; `analysis/microdata_pairwise_refinement.py` computes the information refinements and checks the pairwise formulas against linear programmes. These scripts expect the authorized Stata file in `data/microdata/`.

`analysis/release_design_extension.py` compares the prospective rule, realized pair rankings, direct three-way releases, budget classifications and group gaps. It uses the previously computed state distributions and verifies the single-pair endpoints and worst remaining widths with 1,635 independent linear-programme solves, including 125 boundary and interior marginal vectors. The maximum discrepancy across these checks and reconciliation with previous pairwise results is below 4×10^{-15} . The budget grid is $\{0.5, 1, 1.5, 2\}$ normalized units per adult. Exact marginal ties use the fixed ordering phone, internet, account; zero-width and sign comparisons use a numerical tolerance recorded in the script.

D.4 Device and payment checks

`analysis/indicator_scope_check.py` codes the stricter device measure as `con1==1` and `con9==1`: the respondent owns a phone and reports that the main mobile phone used is a smartphone. Nonowners remain in the denominator with value zero. The 322 don't-know or refusal responses among phone owners in the low- and middle-income sample are also coded zero, following the affirmative-response definition. We restrict comparisons to economies where the question is observed for all phone owners. All 93 baseline economies satisfy this condition, yielding the same 97,923 adults under both bundle definitions.

The official `anydigpayment` indicator records making or receiving a digital payment in the past year. It is observed for every respondent in 98 economies, including 92 of the baseline economies; Russia lacks this module. The payment comparison therefore uses 96,923 respondents, with each conditional rate weighted within its economy and bundle group. Economies with an unavailable module are excluded from this comparison. Official smartphone and payment marginals reconcile in 95 and 98 economies, respectively, with maximum differences below 4×10^{-15} . All-pair smartphone endpoints agree with 190 independent linear-programme solves to within 2×10^{-14} . The outputs retain country-level estimates, coverage checks and source hashes; licensed records remain outside the distributable package.

D.5 An illustration of information releases

Consider the illustrative marginals (0.85, 0.70, 0.65). We construct joint moments from independent Bernoulli variables and progressively release them to an analyst who allows unrestricted dependence. With unit upgrade costs and $f = 1$, marginals alone give the cost interval [1.15, 1.60]. Adding the phone–internet intersection, 0.595, narrows it to [1.205, 1.555]; adding all three pairwise intersections narrows it to [1.3975, 1.4425]. The three-way intersection, 0.38675, point-identifies cost at 1.41325. With all pairwise moments, the remaining interval reflects uncertainty about the three-way delivery-cost term. Section 10 performs the same comparison with observed overlaps.